

Revealing Hidden Alpha in Large-Cap Stocks: LLM-Driven Sentiment Analysis of Japanese 10-K Reports

Moe Nakasuji*¹ Katsuhiko Okada*¹ Yasutomo Tsukioka*¹ Takahiro Yamasaki*²

^{*1}Kwansei Gakuin University ^{*2}Osaka Sangyo University

This study extends prior research on using large language models (LLMs) to uncover return-predictive sentiment in Japanese 10-K reports by focusing on highly liquid stocks. Building on a dataset of Tokyo Stock Exchange-listed firms (2014–2023) and previously established methodologies, we narrow our scope to the TOPIX 100 and TOPIX 500—indices composed of the largest Japanese companies by market capitalization. Despite expectations that these well-followed and actively traded stocks should incorporate public information more efficiently, LLM-derived sentiment still predicts future returns, with larger abnormal returns (alpha) than when all listed stocks are included. These findings highlight the robustness of LLM-based approaches in detecting subtle signals within corporate disclosures and challenge the notion that highly liquid markets fully reflect available information. By highlighting the predictive power of LLM-extracted sentiment in large-cap portfolios, this study offers practical insights into how advanced natural language processing can enhance investment strategies, even in supposedly efficient segments of the equity market.

1. Introduction

The rapid evolution of natural language processing (NLP) technologies, especially large language models (LLMs), has opened new avenues for extracting financial insights from qualitative disclosures such as corporate annual reports. In Okada et al. (2024), applying LLMs like GPT-4 and Claude to Japanese 10-K filings revealed a notable connection between sentiment expressed in Management Discussion and Analysis (MD&A) sections and subsequent stock returns. These findings not only challenged the weak form of the Efficient Market Hypothesis (EMH) but also demonstrated the capacity of advanced NLP techniques to capture signals overlooked by traditional dictionary-based metrics and earlier machine learning methods.

However, one recurring question has been whether these documented predictive signals hold in more efficient market segments—particularly large-cap stocks, which tend to be more liquid, extensively covered by analysts, and thus presumably more efficiently priced. Addressing this gap, our current study narrows the focus to two major subsets of Japanese equities: the TOPIX 100 and TOPIX 500 indices, each composed of the largest firms listed on the Tokyo Stock Exchange. By restricting our trading universe to highly liquid stocks, we aim to determine whether LLM-derived sentiment retains its predictive power amid lower transaction costs, tighter bid-ask spreads, and intense investor scrutiny.

Our new empirical findings are striking despite expecting a diminished effect in well-followed and actively traded stocks, the sentiment-based long–short strategies for both TOPIX 100 and TOPIX 500 generate statistically significant alpha. Notably, we find that these large-cap portfolios exhibit even stronger abnormal returns than the broader set of listed firms. This raises intriguing questions about the continued presence of mispriced sentiment, even in segments of the market typically considered more efficient.

In the sections that follow, we detail our data and methodology—closely aligning with our previous approach but focusing on large-cap stocks—and present our empirical results. Lastly, we discuss the implications of these findings for asset pricing, market efficiency, and practical investment strategies.

2. Data and Methodology

2.1 Data

The 10-K reports analyzed in this study were sourced from the Electronic Disclosure for Investors’ NETwork, operated by Japan’s Financial Services Agency. We focused on firms whose fiscal year ends in March, the standard for most TSE-listed companies, resulting in 16,363 firm-year observations and over 90 million words of textual data. This extensive dataset enables a comprehensive analysis of management sentiment across the Japanese market.

Table 1 summarizes the dataset, including the number of firms (10-K reports) per year, the total stocks analyzed, and the average length of the MD&A sections in characters. The MD&A section, equivalent to its U.S. counterpart, offers rich qualitative insights into financial condition, operational results, and management outlook, making it a critical focus for evaluating sentiment and investor perceptions.

Table 1 : Summary of Dataset Characteristics: 10-K Reports from Firms Listed on the Tokyo Stock Exchange (2014–2023)

	Number of firms	Average length of MD&A (Number of characters)	Min	25%	Median	75%	Max
2014	1,012	6,159	1,910	4,409	5,339	6,654	43,996
2015	1,052	6,136	2,230	4,384	5,365	6,764	43,943
2016	1,069	6,095	1,956	4,414	5,412	6,821	40,281
2017	1,091	5,829	2,094	4,290	5,149	6,491	32,411
2018	1,165	6,294	2,291	4,720	5,751	7,106	29,553
2019	1,190	6,510	2,233	4,896	5,960	7,289	29,542
2020	1,188	7,501	2,966	5,633	6,834	8,536	29,622
2021	1,191	7,345	2,855	5,537	6,639	8,233	29,576
2022	1,176	7,278	2,663	5,506	6,630	8,251	29,614
2023	1,001	7,210	2,797	5,497	6,607	8,117	29,558
Total	11,135	6,636	1,910	4,899	5,052	7,575	43,996

Note: Table 1 presents breakdown of the dataset used in this study, summarizing the annual number of firms listed on the Tokyo Stock Exchange, their corresponding 10-K reports, and the total stocks analyzed. A particular focus is placed on the Management Discussion and Analysis (MD&A) sections. This section, equivalent to the MD&A in U.S. filings, provides in-depth qualitative narratives on financial conditions, operational results, and cash flow status.

Contact: Moe Nakasuji, Kwansei Gakuin University,
Nakasuji@kwansei.ac.jp

2.2 Sentiment Extraction Method

To extract sentiment from the MD&A sections, we utilize five approaches: a financial polarity dictionary for Japanese financial texts, DeBERTaV2, and three large language models (LLMs)—GPT-4, Claude, and Gemini.

Dictionary-Based Methods

Using the Financial Polarity Dictionary, we compute two metrics:

Tone Ratio: Measures the balance of positive and negative words:

$$\text{Tone Ratio} = \frac{N^+ - N^-}{N^+ + N^-}$$

where N^+ and N^- denote the number of positive and negative words, respectively. **Tone Score:** Captures sentiment intensity by summing scores assigned to each word:

$$\text{Tone Score} = \sum_{i=1}^n s_i \quad s_i \in \mathbb{R}$$

where s_i is the sentiment score of the i -th word, and n is the total number of words in the MD&A section.

DeBERTaV2 Model

The DeBERTaV2 model uses a combined objective for masked language modeling (MLM) and sentiment classification:

$$L = L_{MLM} + \lambda L_{SC}$$

where

$$L_{MLM} = -\frac{1}{|M|} \sum_{i \in M} \log P(x_i | X_{\setminus i})$$

$$L_{SC} = -\frac{1}{N} \sum_{j=1}^N \sum_{c \in C} y_{j,c} \log P(c | X_j)$$

Here, M is the set of masked tokens, N is the number of samples, C represents sentiment classes, $y_{j,c}$ is the ground-truth label for class c , and λ is a hyperparameter. The final sentiment score for a document D is:

$$S(D) = \sum_{j=1}^m w_j \cdot \sum_{c \in C} s(c) \cdot P(c | X_j)$$

where w_j is the weight for sentence j , and $s(c)$ is the numerical sentiment score for class c .

Large Language Models (LLMs)

For GPT-4, Claude, and Gemini, we develop standardized Japanese prompts instructing models to calculate positive and negative sentiment scores summing to 100. The prompts ensure consistent evaluation across MD&A sections. Each LLM processes the text using its unique architecture:

- GPT-4 emphasizes adaptability through task-specific prompts.
- Claude focuses on logical consistency and contextual depth.
- Gemini integrates semantic and thematic context dynamically.

These methodologies provide a comparative framework to evaluate the efficacy of traditional dictionary-based metrics and advanced LLMs in capturing management sentiment and its implications for future stock returns. Table 2 summarizes the key characteristics of these measures, offering insights into their distribution, central tendencies, and variability.

Table 2 Distribution of Sentiment Scores Extracted from MD&A Sections Using Five Different Methods

	Mean	Min	25%	Median	75%	Max
Tone Ratio	-0.144	-0.393	-0.184	-0.142	-0.102	0.083
Tone Score	-3.265	-194.445	-11.188	-1.872	6.108	92.435
DeBERTaV2	0.689	0.004	0.182	0.998	0.999	0.999
GPT-4	0.575	0.100	0.400	0.600	0.700	0.900
Claude	0.616	0.000	0.500	0.600	0.700	0.800
Gemini	0.517	0.100	0.400	0.600	0.700	0.900

Note: Sentiment scores are calculated using six measures from five methodologies. The first two, Tone Ratio and Tone Score, are based on the Financial Polarity Dictionary by the University of Tokyo. Tone Ratio measures the balance of positive (N^+) and negative (N^-) words while Tone Score aggregates sentiment strength of each word. The third measure uses the DeBERTaV2 model, fine-tuned on the chABSA dataset, to compute sentiment probabilities and document-level scores. The final three measures employ GPT-4-mini, Claude 3 Haiku, and Gemini 1.5 Flash, which generate sentiment scores scaled to sum to 1 based on standardized prompts in Japanese.

3. Results

3.1 Sentiment-Based Long-Short Portfolio Return

The primary aim of this study is to further examine whether large language models (LLMs) can uncover predictive signals in Japanese 10-K reports when we restrict the trading universe to more liquid stocks. Prior research indicates that small-cap segments may offer greater inefficiencies; however, large-cap stocks typically feature lower transaction costs, tighter bid-ask spreads, and higher analyst coverage, all of which should, in theory, reduce the potential for sentiment-driven mispricing. To investigate this, we narrowed our focus to two subsets of companies: the TOPIX 500, comprising the top 500 firms listed on the Tokyo Stock Exchange by market value, and the TOPIX 100, an even more exclusive group of large-cap stocks known for their high liquidity and ease of short-selling.

Using the same methodology described in our previous analysis—sorting firms into quintiles based on sentiment scores extracted from MD&A sections—we construct value-weighted long–short portfolios. The long leg consists of stocks with the highest sentiment scores, and the short leg consists of those with the lowest sentiment, measured at the end of June each year. The portfolios are then held from July of year through June of year $t+1$. This procedure is repeated annually from 2014 through 2023 for each of the six sentiment extraction methods (Tone Ratio, Tone Score, DeBERTaV2, GPT-4, Claude, and Gemini).

Interestingly, even within these more liquid segments, the sentiment-based portfolios showed meaningful predictive power. Figure 1 indicates that the cumulative returns on the long–short strategies derived from GPT-4 and Claude sentiment measures—focused specifically on the TOPIX 100 universe—maintained a pronounced pattern of negative returns, echoing our earlier findings with broader sets of stocks. Given that the TOPIX 100 comprises the most liquid and heavily traded firms on the Tokyo Stock Exchange, these abnormal returns are particularly noteworthy, suggesting that even under conditions of minimal transaction costs and ready shorting opportunities, sentiment-based mispricing persists. Such results highlight the practical feasibility of these strategies, as their implementation in highly liquid markets makes them not only theoretically interesting but

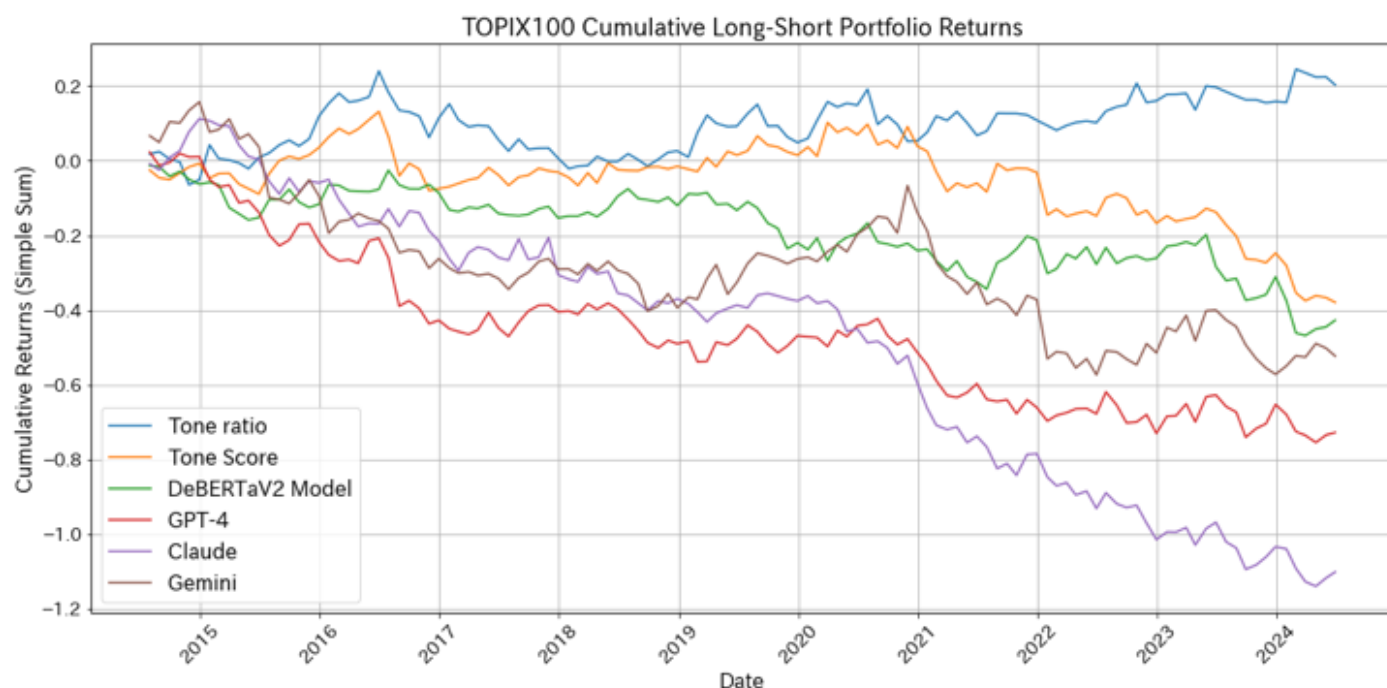


Figure 1: Long-Short Calendar Time Portfolio Performance Based on Various Sentiment Indicators

also potentially actionable for active portfolio managers. Equally notable, while Tone Ratio, Tone Score, and DeBERTaV2 continued to exhibit only mild or negative predictive performance, the portfolios constructed using Gemini demonstrated moderate yet still significant capacity to anticipate future returns.

3.2 Performance Evaluation Across Different Market Segments

Our previous research shows that sentiment-based long-short portfolios using GPT-4, Claude, and Gemini consistently generated significant abnormal returns (alphas) when applied to all stocks in the TSE Prime market. The present study refines this approach by limiting the universe to two more liquid subsets: the TOPIX 500 and the TOPIX 100. Table 3, which focuses on the LLM-based methods, reports the results of time-series regressions (Fama-French three-factor (FF3), Carhart four-factor (FFC4), and Fama-French five-factor (FF5)) testing whether the monthly excess returns on these long-short portfolios can be explained by standard risk factors in these large-cap segments. Notably, the dictionary-based methods—Tone Ratio and Tone Score—and the DeBERTaV2 model are excluded from Table 3 because none of these approaches generated statistically significant alphas in any subset of the data.

Consistent with Okada et al. (2024) broader sample findings, the alpha estimates for GPT-4, Claude and Gemini remain both economically and statistically significant within the TOPIX 500 and TOPIX 100, but the magnitude of these alphas mostly increases as we move to larger and more liquid stocks. For example, GPT-4 exhibits a monthly alpha of -0.50% when using all TSE Prime stocks, compared with -0.29% in the TOPIX 500 and a more pronounced -0.58% in the TOPIX 100. Similarly, Claude achieves a monthly alpha of -0.76% for all stocks, versus -0.80% in the TOPIX 500 and -0.82% in the TOPIX 100. Although Gemini's alpha is somewhat smaller, it likewise grows in these

more liquid subsets, moving from -0.41% to -0.50% and -0.61% in the TOPIX 500 and TOPIX 100, respectively.

By contrast, none of the dictionary-based (“bag of words”) approaches—Tone Ratio and Tone Score—nor the more traditional NLP model DeBERTaV2 produce consistently significant alphas in any subset of the data. These methods fail to identify sentiment cues that lead to persistent mispricing, suggesting that purely counting positive and negative terms or relying on less sophisticated deep learning architectures does not sufficiently capture the nuanced language in MD&A sections.

4. Conclusion

This study builds on previous work (Okada et al., 2024) by investigating whether the predictive power of large language model (LLM)-derived sentiment, observed in broad Japanese equity markets, remains robust within highly liquid stocks. Focusing on the TOPIX 500 and the TOPIX 100, we find that the abnormal returns (alphas) generated by GPT-4, Claude, and to a lesser extent Gemini, are not only statistically significant but also exceed those documented in more comprehensive market samples. These results are especially striking given that larger, more liquid firms typically attract extensive analyst coverage and are presumed to incorporate public information more efficiently.

By contrast, dictionary-based approaches (Tone Ratio and Tone Score) and the DeBERTaV2 model continue to produce minimal or insignificant alphas, reinforcing the notion that simpler text-based methods struggle to capture the nuances present in 10-K MD&A sections. The heightened alpha in large-cap segments is particularly noteworthy, challenging the conventional assumption that institutional scrutiny and high turnover minimize mispricing in blue-chip stocks. Instead, our findings suggest that advanced LLMs can uncover subtle sentiment-related signals—signals that

Table 3 Regression Results for Long-Short Portfolios with Different Formation Strategies Based on Sentiment Scores

Panel A (TSE Prime)									
	GPT-4			Claude			Gemini		
	FF3	FF4	FF5	FF3	FF4	FF5	FF3	FF4	FF5
α	-0.0050 **	-0.0054 **	-0.0052 **	-0.0076 ***	-0.0083 ***	-0.0079 ***	-0.0041	-0.0046	-0.0036
Rm-Rf	0.0883	0.1352 **	0.0031	-0.0002	0.0650	-0.0767	0.1666 **	0.2184 ***	0.0854
SMB	-0.0936	-0.1472	-0.0803	-0.1828	-0.2573 *	-0.1707	-0.2564 *	-0.3155 **	-0.2194
HML	-0.5233 ***	-0.4611 ***	-0.1854	-0.6021 ***	-0.5155 ***	-0.2586 *	-0.8214 ***	-0.7527 ***	-0.5591 ***
WML		0.1921 *			0.2672			0.2119 *	
RMW			0.5781 **			0.5376 *			0.1164
CMA			-0.1864			-0.1940			-0.5380 **
Adj R-Squared	0.2485	0.2646	0.2912	0.2187	0.2458	0.2337	0.3774	0.3884	0.3845
Panel B (TOPIX 500)									
α	-0.0029	-0.0033	-0.0028	-0.0080 ***	-0.0086 **	-0.0083 ***	-0.0050 *	-0.0054 *	-0.0047 *
Rm-Rf	0.1078 *	0.1498 **	0.0367	-0.0088	0.0517	-0.0837	0.1960 ***	0.2422 **	0.0945
SMB	-0.2261 *	-0.2741 **	-0.2133 *	-0.1808	-0.2499 *	-0.1584	-0.2440 *	-0.2967 **	-0.1988
HML	-0.5718 ***	-0.5161 ***	-0.3105 **	-0.5722 ***	-0.4920 ***	-0.2315 *	-0.8092 ***	-0.7479 ***	-0.4720 ***
WML		0.1721 *			0.2477 **			0.1892	
RMW			0.2305			0.5524 **			0.2956
CMA			-0.3498 *			-0.1680			-0.5195 **
Adj R-Squared	0.2920	0.3039	0.2953	0.2123	0.2371	0.2315	0.3944	0.4026	0.4153
Panel C (TOPIX 100)									
α	-0.0058 **	-0.0058 ***	-0.0062 **	-0.0082 ***	-0.0083 ***	-0.0089 ***	-0.0061 *	-0.0061 *	-0.0062 *
Rm-Rf	0.1057	0.1116	0.0433	-0.0159	-0.0018	-0.0916	0.2452 ***	0.2452 **	0.1464
SMB	-0.2136	-0.2203	-0.1822	-0.2094	-0.2255	-0.1778	-0.1845	-0.1844	-0.1037
HML	-0.5331 ***	-0.5253 ***	-0.1884	-0.4522 ***	-0.4335 ***	-0.0621 *	-0.6842 ***	-0.6843 ***	-0.3246 *
WML		0.0240			0.0577			-0.0003	
RMW			0.6275 **			0.7700 ***			0.4784
CMA			-0.0508			-0.0177			-0.3371
Adj R-Squared	0.2110	0.2040	0.2183	0.1163	0.1099	0.1629	0.2484	0.2415	0.2501

This table presents the results of time-series regressions of the monthly excess returns on value-weighted long–short portfolios constructed using LLM-based sentiment scores (GPT-4, Claude, and Gemini) extracted from the MD&A sections of Japanese 10-K reports. The dictionary-based methods (Tone Ratio and Tone Score) and the DeBERTaV2 model are excluded, as they did not yield significant abnormal returns. In each panel, long–short portfolios are formed by taking long positions in the top quintile and short positions in the bottom quintile of stocks, based on their sentiment scores. These portfolios are held from July of year t through June of year $t+1$. Panel A reports results for all TSE Prime stocks, Panel B for the TOPIX 500 subset, and Panel C for the TOPIX 100 subset. The regression models employed are the Fama–French three-factor model (FF3), the Carhart four-factor model (FFC4), and the Fama–French five-factor model (FF5). The reported alphas represent the intercepts of these regressions, indicating abnormal returns not explained by the included risk factors. Statistical significance at the 1%, 5%, and 10% levels is denoted by ***, **, and *, respectively.

appear to remain underappreciated even in these supposedly more efficient market segments.

From a practical perspective, the results highlight the potential profitability of deploying LLM-driven sentiment measures in real-world investment strategies focusing on highly liquid equities. Future research may investigate how further advancements in model architectures or training processes could enhance the extraction of predictive features from financial texts. Additionally, extending these analyses to other international markets or incorporating event-specific windows (e.g., earnings announcements) could offer deeper insights into the dynamics of sentiment-based mispricing. Overall, our study underscores the continued importance of sophisticated NLP techniques in uncovering alpha opportunities, thereby questioning the scope of market efficiency—even within large-cap universes.

Acknowledgement

This research was supported by the Grant-in-Aid for Scientific Research (B), Project Number 24K00298.

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